

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2345

Fairfield Washington 138 kV
March 19, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2345. The Interconnection Customer proposes a 137.6 MW solar plus storage facility interconnecting at the Fairfield Washington 138 kV substation in the Pennsylvania area. The request was received on August 28, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 460 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,187,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WYMAN-GORDON to ONTELAUNEE ENERGY CENTER 138kV	ELIZA CRK to ONTELAUNEE ENERGY CTR 345kV	124.7	264	19.6
WYMAN-GORDON to TRISONIC SUBSTATION 138kV	ELIZA CRK to ONTELAUNEE ENERGY CENTER 345kV	122.7	253	19.6
REDONDO to LEAKE 138kV	ELIZA CRK to ONTELAUNEE ENERGY CENTER 345kV	117.9	225	23.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WYMAN-GORDON to ONTELAUNEE ENERGY CTR 138kV	Rebuild line with higher-capacity conductor	\$2,598,000	100% (\$2,598,000)
WYMAN-GORDON to TRISONIC SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$1,037,000	100% (\$1,037,000)
LEAKE to REDONDO 138kV	Rebuild line with higher-capacity conductor	\$1,552,000	100% (\$1,552,000)
	Total allocated to project		\$5,187,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.